

Practice Questions for 2024-09-05-SimplexEfficiency

Generated by Scribe.AI

December 17, 2024

1 Questions

1. According to Dantzig, for a linear programming problem with $m < 50$ and $m + n < 200$, what is the typical number of iterations the simplex method requires?

- A. Approximately $3m$
- B. Usually less than or equal to $\frac{3m}{2}$
- C. Always $2^n - 1$
- D. Always n
- E. Always $m + n$

2. What is the worst-case number of steps required by the simplex method using the Largest Coefficient Rule, as demonstrated by the Klee-Minty example?

- A. n
- B. $\frac{3m}{2}$
- C. $3m$
- D. $2^n - 1$
- E. $m+n$

3. In the Klee-Minty worst-case example, what is the number of steps required by the simplex method using the Smallest Coefficient Rule?

- A. $2^n - 1$
- B. $\frac{3m}{2}$
- C. $3m$
- D. n
- E. $m+n$

2 Answers

1. According to Dantzig, for a linear programming problem with $m < 50$ and $m + n < 200$, what is the typical number of iterations the simplex method requires?
 - A. Answer A is incorrect because while the number of iterations can rarely approach $3m$, it is not the typical case.
 - B. Answer B is correct because, according to the slide, the number of iterations is usually less than or equal to $\frac{3m}{2}$.
 - C. Answer C is incorrect because $2^n - 1$ is the number of steps for the largest coefficient rule in the Klee-Minty worst-case example, not the typical case.
 - D. Answer D is incorrect because n is the number of steps for the smallest coefficient rule in the Klee-Minty worst-case example, not the typical case.
 - E. Answer E is incorrect because the number of iterations is not typically $m + n$.
2. What is the worst-case number of steps required by the simplex method using the Largest Coefficient Rule, as demonstrated by the Klee-Minty example?
 - A. Answer A is incorrect because n is the number of steps for the Smallest Coefficient Rule in the Klee-Minty example.
 - B. Answer B is incorrect because $\frac{3m}{2}$ is the typical number of iterations according to Dantzig, not the worst-case for the Largest Coefficient Rule.
 - C. Answer C is incorrect because $3m$ is a rare upper bound on the number of iterations according to Dantzig, not the worst-case for the Largest Coefficient Rule.
 - D. Answer D is correct because the slide states that the Largest Coefficient Rule requires $2^n - 1$ steps in the Klee-Minty example.
 - E. Answer E is incorrect because $m + n$ is not the number of steps for the Largest Coefficient Rule in the Klee-Minty example.
3. In the Klee-Minty worst-case example, what is the number of steps required by the simplex method using the Smallest Coefficient Rule?
 - A. Answer A is incorrect because $2^n - 1$ is the number of steps for the Largest Coefficient Rule in the Klee-Minty example.
 - B. Answer B is incorrect because $\frac{3m}{2}$ is the typical number of iterations according to Dantzig, not the worst-case for the Smallest Coefficient Rule.
 - C. Answer C is incorrect because $3m$ is a rare upper bound on the number of iterations according to Dantzig, not the worst-case for the Smallest Coefficient Rule.
 - D. Answer D is correct because the slide states that the Smallest Coefficient Rule requires n steps in the Klee-Minty example.
 - E. Answer E is incorrect because $m + n$ is not the number of steps for the Smallest Coefficient Rule in the Klee-Minty example.